

AWS Cloud Architecture &

Infrastructure Design Document

TrueNorth — Hostel Management Platform

Version 1.6

Field	Detail
Document Version	v1.6
Prepared By	Zenxvio Internal Team
Date	July 13, 2026
Classification	Internal Review
Status	Working Draft
Region	ap-south-1 (Mumbai, India)

[!NOTE] **Working Draft:** This document is actively maintained and subject to revision as the platform evolves. All revenue and infrastructure cost figures in Section 14 are hypothetical projections for planning purposes only. Actual pricing strategy is not fixed.

Table of Contents

1. Introduction
2. Complete System Architecture
3. Dual-Environment Strategy (Staging & Production)
4. Network Architecture & VPC Security
5. Frontend Architecture & State Management
6. Backend Architecture
7. Full Database Schema & Architecture
8. Cloud Storage & Document Management
9. Identity, Access Management & Secrets
10. CI/CD Pipeline & Deployment
11. Observability & Monitoring
12. Disaster Recovery & Backup
13. Cost Optimization & MVP Budget
14. Revenue & Infrastructure Forecast (*Hypothetical*)
15. Exhaustive API Documentation

1. Introduction

1.1 Purpose

The TrueNorth Hostel Management Platform is a comprehensive, multi-tenant SaaS portal designed to streamline daily operations for Admins, Wardens, and Tenants across multiple hostel properties. This document serves as the authoritative engineering reference for the platform's complete AWS cloud infrastructure, database schema, deployment strategy, security posture, and financial forecasting.

1.2 Scope

This document covers the complete architectural footprint across two deliberate deployment phases:

- **Phase 1 (MVP — Current):** Single EC2 t3.micro on AWS Free Tier (ap-south-1), serving up to 2,000 registered users across 30 hostels at ~₹42/month.
- **Phase 2 (Scale — Future):** Migration to ECS Fargate with Auto-Scaling, Multi-AZ RDS, and ElastiCache — triggered only when the platform exceeds 30 active hostels.

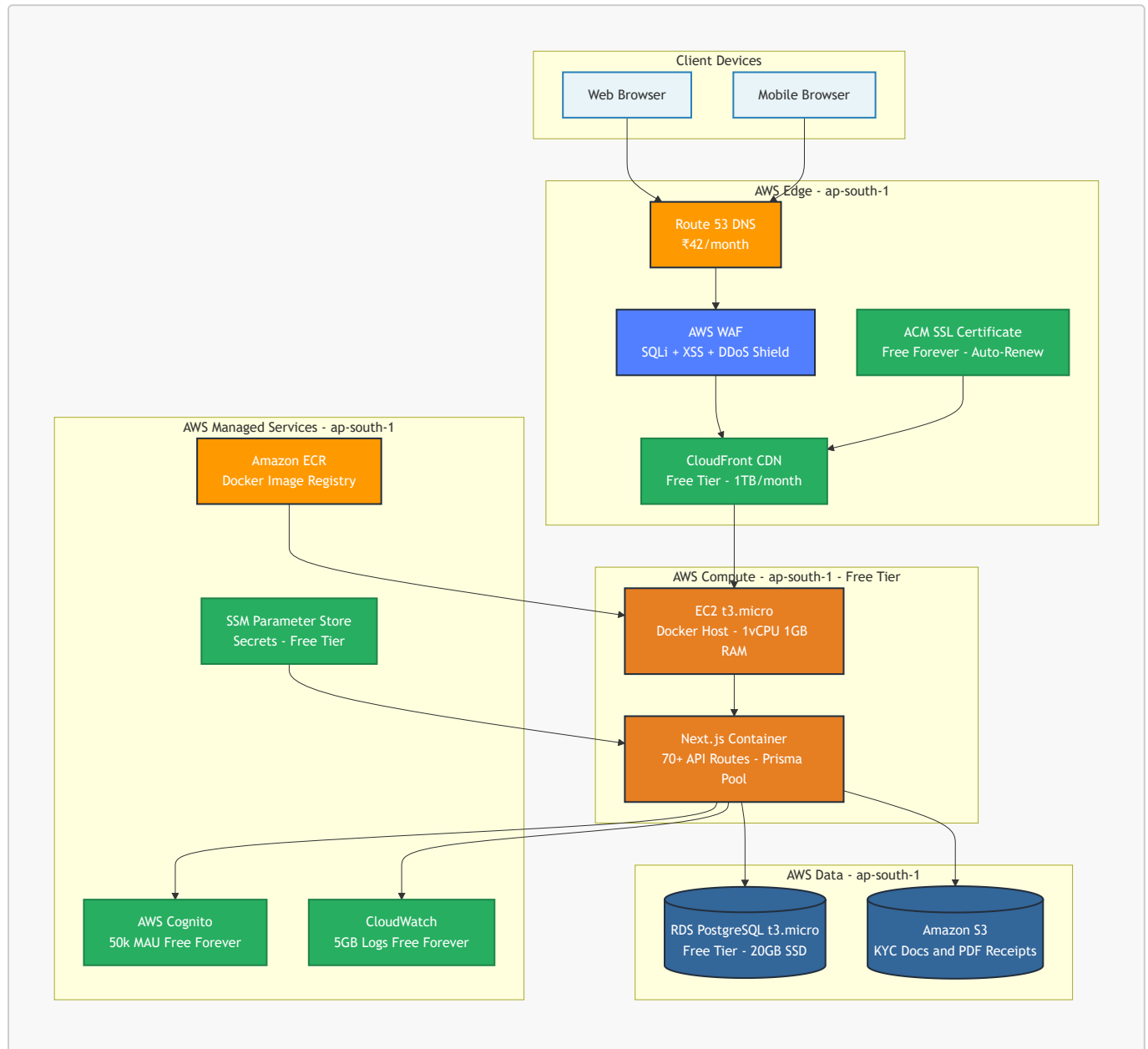
1.3 Definitions & Acronyms

Acronym	Definition	Context in Platform
AWS	Amazon Web Services	Primary cloud provider (Mumbai region)
EC2	Elastic Compute Cloud	Phase 1 virtual server hosting the Docker container
ECS	Elastic Container Service	Phase 2 managed container orchestration
RDS	Relational Database Service	Managed PostgreSQL in ap-south-1
ECR	Elastic Container Registry	Docker image registry used by CI/CD pipeline
S3	Simple Storage Service	Object storage for KYC docs and PDF receipts
ACM	AWS Certificate Manager	Free auto-renewing SSL certificate
WAF	Web Application Firewall	Edge protection against SQLi, XSS, DDoS
SSM	Systems Manager Parameter Store	Secure secrets & environment variable storage
JWT	JSON Web Token	Auth token issued by AWS Cognito
ALB	Application Load Balancer	Phase 2 only — not in MVP
VPC	Virtual Private Cloud	Isolated AWS network in ap-south-1
PITR	Point-In-Time Recovery	RDS backup restore to any second in 35 days
MAU	Monthly Active Users	Cognito billing unit (50k free permanently)
CDN	Content Delivery Network	CloudFront caches static assets at edge

2. Complete System Architecture

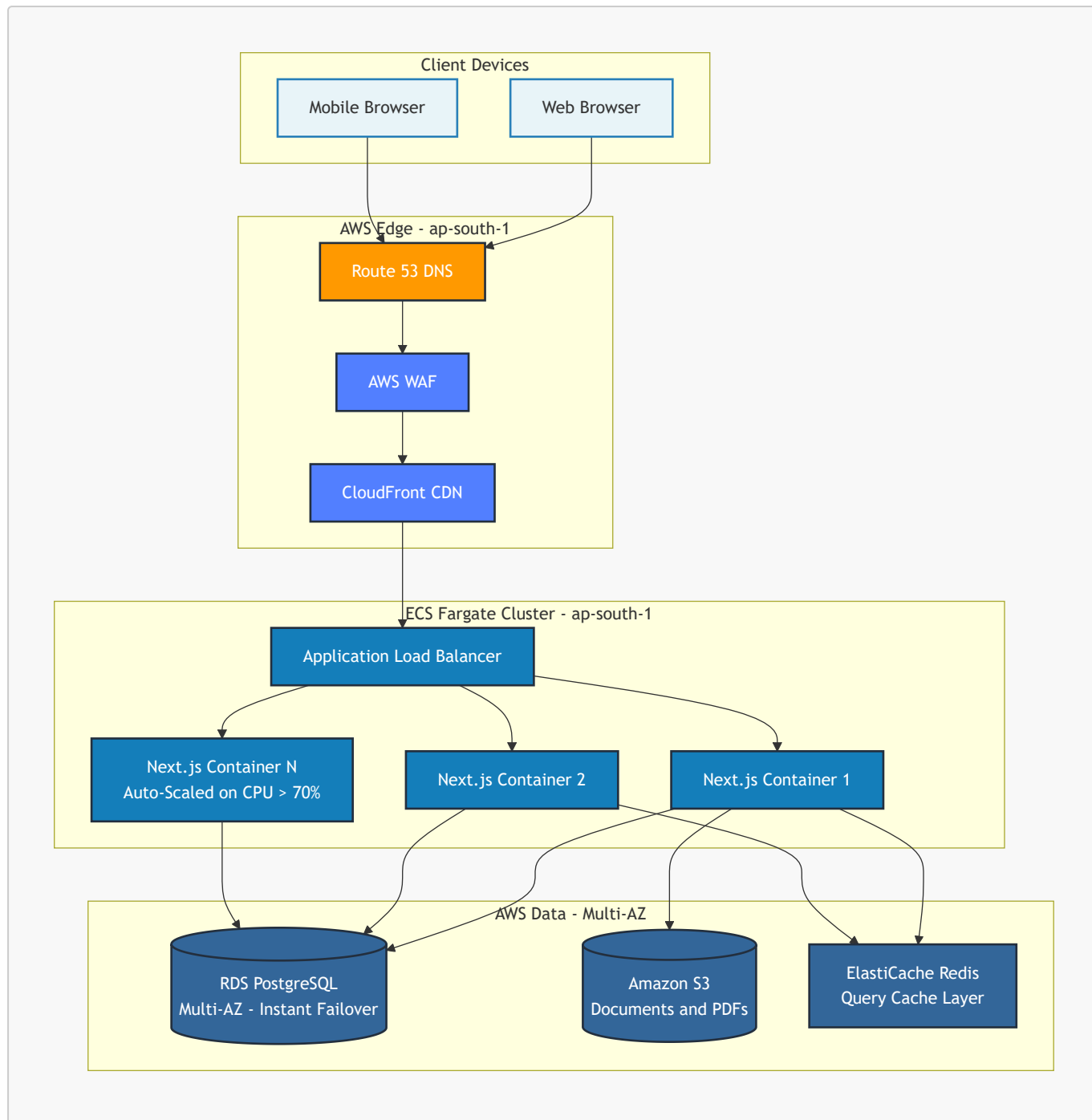
2.1 Phase 1 — MVP Architecture (Current, ap-south-1)

The MVP runs entirely on the AWS Free Tier in the Mumbai region. A single EC2 t3.micro instance runs the Docker container which hosts the entire Next.js application — all 70+ API routes, the frontend renderer, and the Prisma ORM connection pool — as a unified process.



2.2 Phase 2 — Scale Architecture (Future, 30+ Hostels)

Phase 2 is triggered when the platform grows beyond 30 active hostels. The same Docker container image is migrated from EC2 to ECS Fargate for managed auto-scaling. An Application Load Balancer distributes traffic across multiple container replicas.

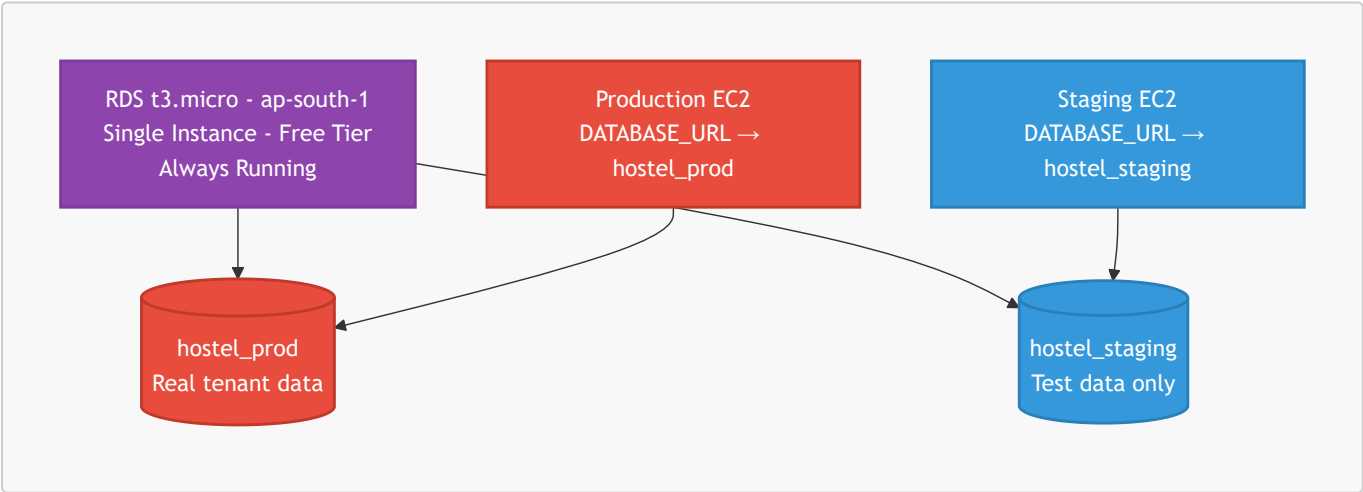


3. Dual-Environment Strategy

3.1 Environment Comparison

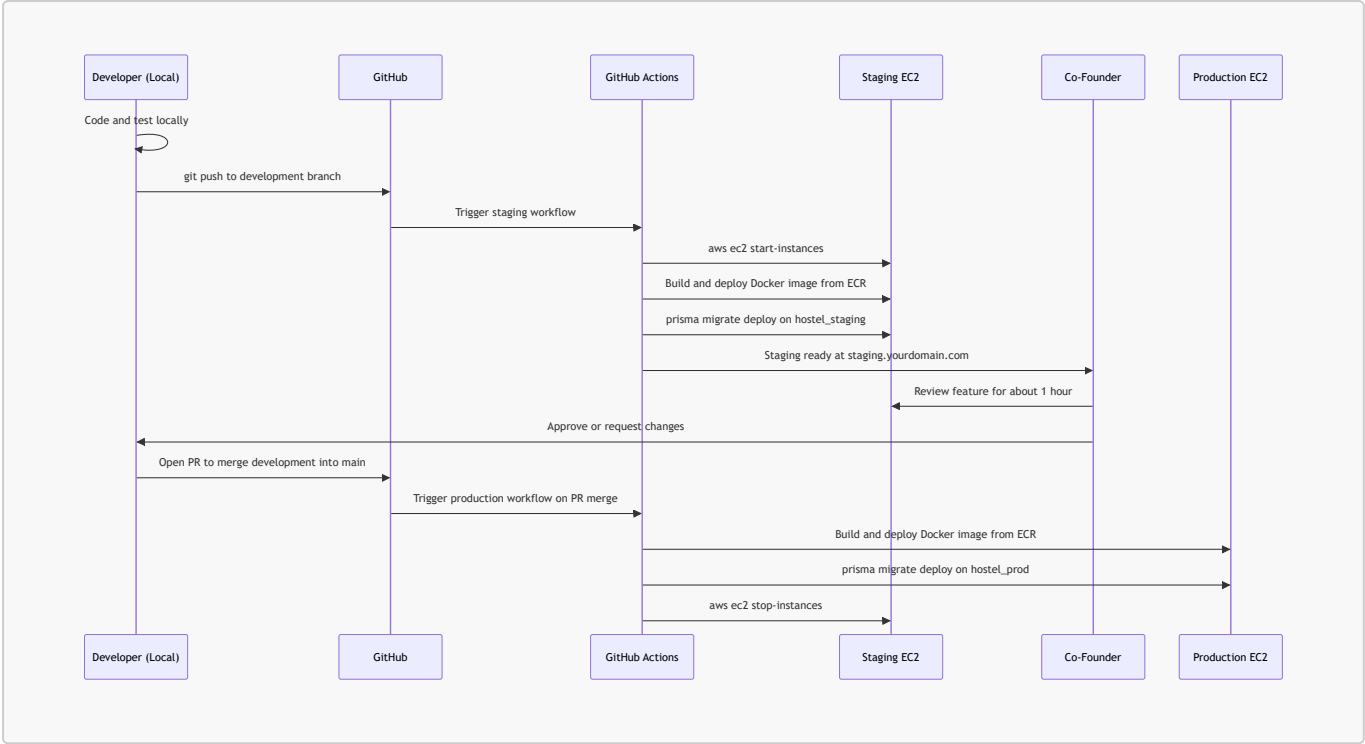
Component	Staging	Production
Purpose	Feature testing & co-founder review	Live system — real tenants
Region	ap-south-1 (Mumbai)	ap-south-1 (Mumbai)
Server	EC2 t3.micro (off when idle)	EC2 t3.micro (24/7)
Database	hostel_staging on shared RDS	hostel_prod on shared RDS
S3 Bucket	hostel-staging-documents	hostel-prod-documents
Cognito Pool	Separate pool (test users only)	Separate pool (real users)
Domain	staging.yourdomain.com	app.yourdomain.com
SSL	ACM (free, auto-renew)	ACM (free, auto-renew)
Deploys from	development branch	main branch via PR only
Runtime/month	~20 hrs (on-demand only)	720 hrs (always on)
Monthly Cost	₹0	₹0

3.2 Shared RDS — Two Isolated Databases



The `DATABASE_URL` secret in SSM Parameter Store is completely different per environment. A staging deployment cannot physically access production data.

3.3 Developer Workflow Sequence

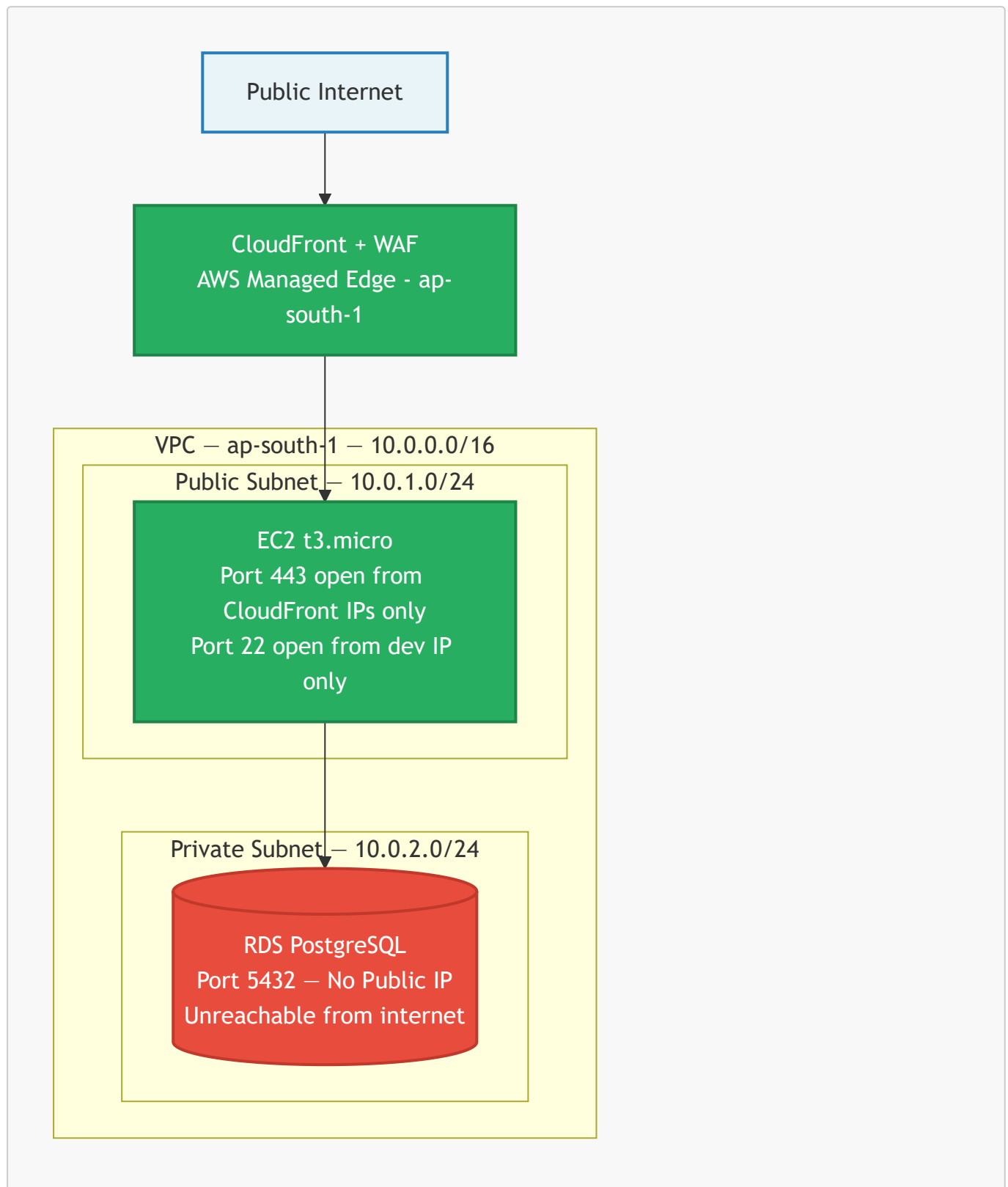


3.4 Free Tier Hours — Stays Under Limit

Instance	Runtime	Hours/Month
Production EC2 (24/7)	Always on	720 hrs
Staging EC2 (on-demand ~1hr/session)	~20 sessions	20 hrs
Combined Total		740 hrs
AWS Free Tier Limit (shared)		750 hrs 
Monthly Cost		₹0

4. Network Architecture & VPC Security

4.1 VPC Network Topology — ap-south-1



4.2 Security Group Rules

EC2 Security Group (sg-app)

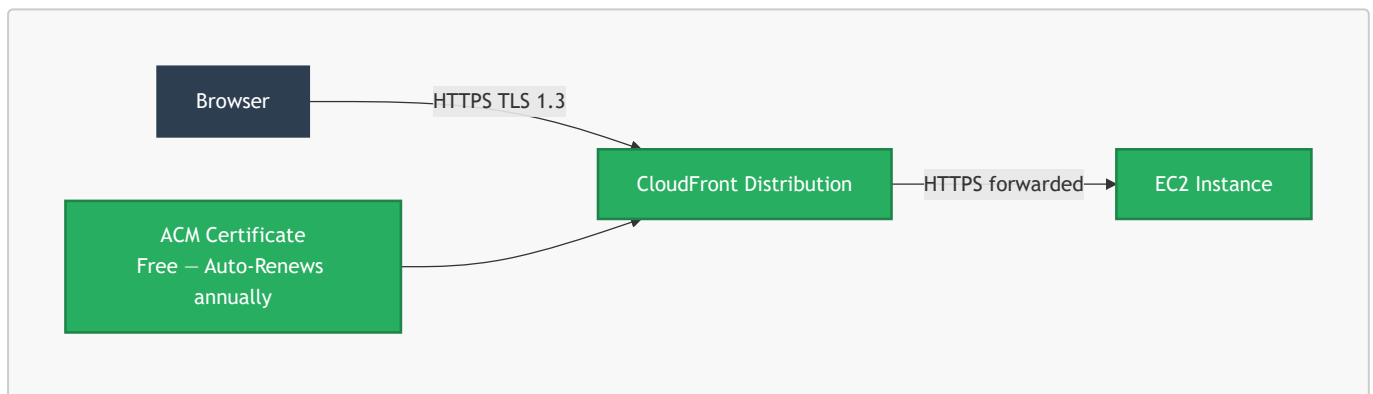
Direction	Protocol	Port	Source	Purpose
Inbound	HTTPS	443	CloudFront managed prefix list	App traffic only
Inbound	SSH	22	Developer static IP	Deployments
Outbound	PostgreSQL	5432	sg-rds (RDS Security Group)	DB queries
Outbound	HTTPS	443	0.0.0.0/0	ECR, SSM, Cognito, CloudWatch

RDS Security Group (sg-rds)

Direction	Protocol	Port	Source	Purpose
Inbound	PostgreSQL	5432	sg-app only	App DB queries
Outbound	None	—	—	No outbound access

[!IMPORTANT] The RDS instance has zero public IP address. It is physically unreachable from the internet. The only network path into the database is from the EC2 instance through the internal VPC routing.

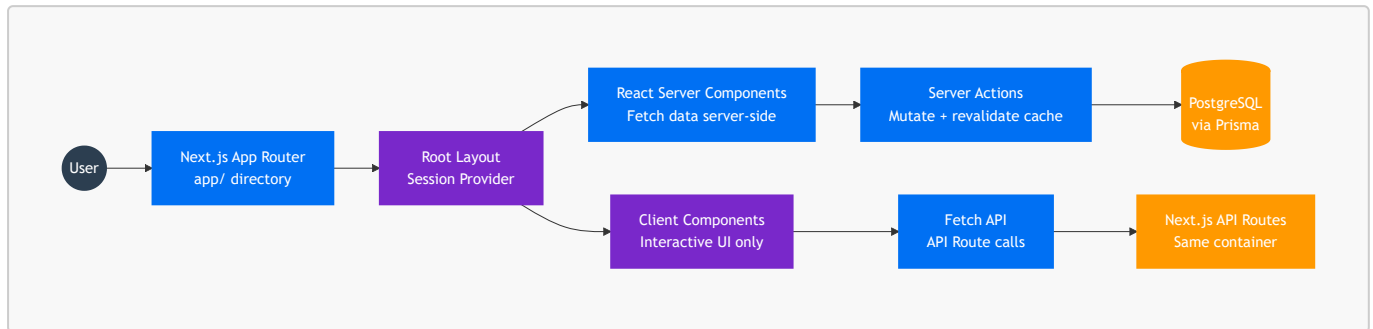
4.3 HTTPS / SSL — AWS Certificate Manager (Free)



ACM issues a wildcard certificate covering both app.yourdomain.com and staging.yourdomain.com. Zero manual renewal — AWS renews it automatically 60 days before expiry.

5. Frontend Architecture & State Management

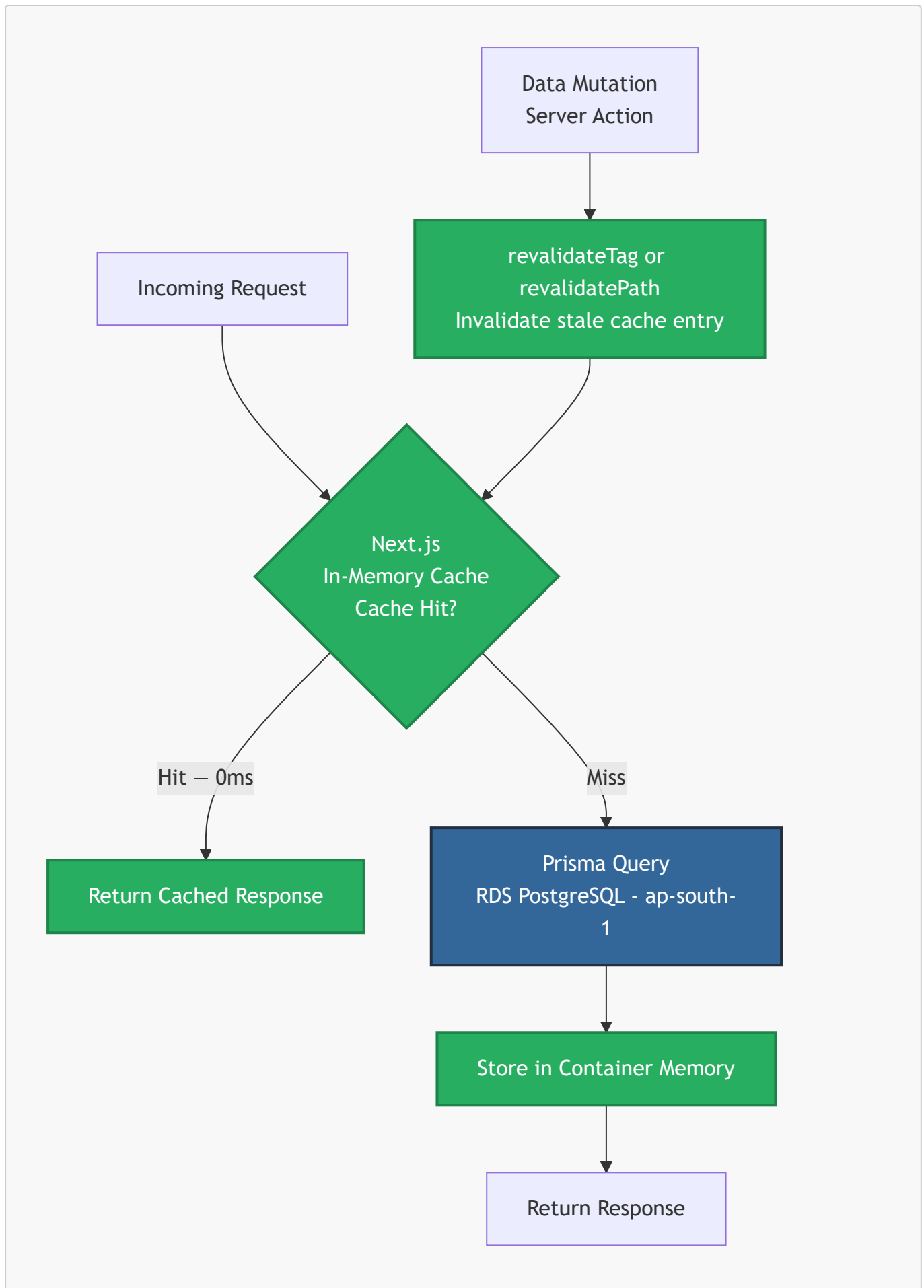
5.1 Frontend Data Flow



5.2 Why We Rejected Redux / Zustand

- **Server Components** fetch data directly from Prisma on the server — zero API call, zero Redux store needed.
- **Server Actions** handle all mutations and call `revalidatePath()` to instantly refresh the UI without a full reload.
- **Client Components** are reserved strictly for interactive elements — not data fetching.

5.3 Next.js Native \$0 Caching (No Redis Needed)

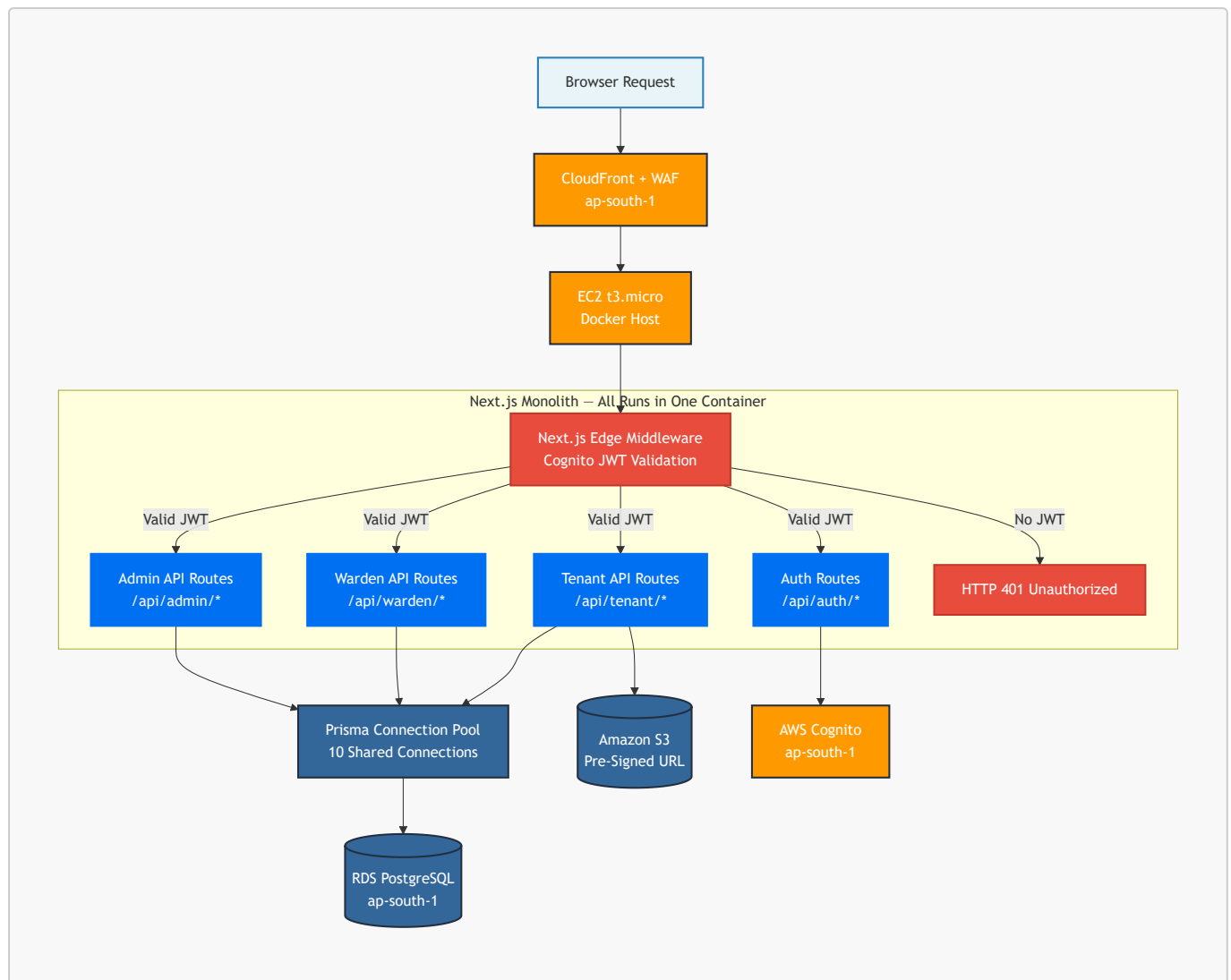


Dashboard reads (available beds, hostel stats, tenant lists) are cached in the container's memory. All wardens in a hostel share the same cached response until a mutation invalidates it. This is functionally equivalent to

Redis for our read patterns — at ₹0 cost. ElastiCache is planned only for Phase 2 at 30+ hostels.

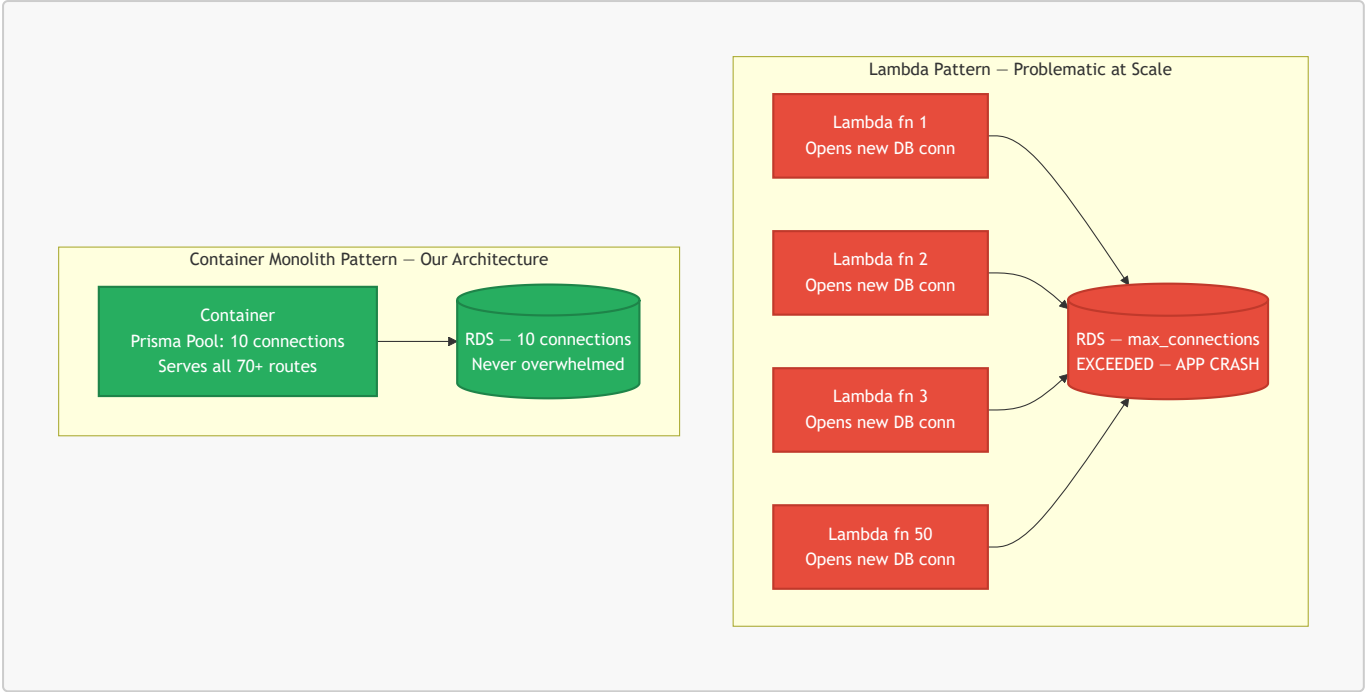
6. Backend Architecture

6.1 Backend Service Flow — All 70+ Routes in One Container



6.2 Why Container Monolith Over Lambda (Architectural Justification)

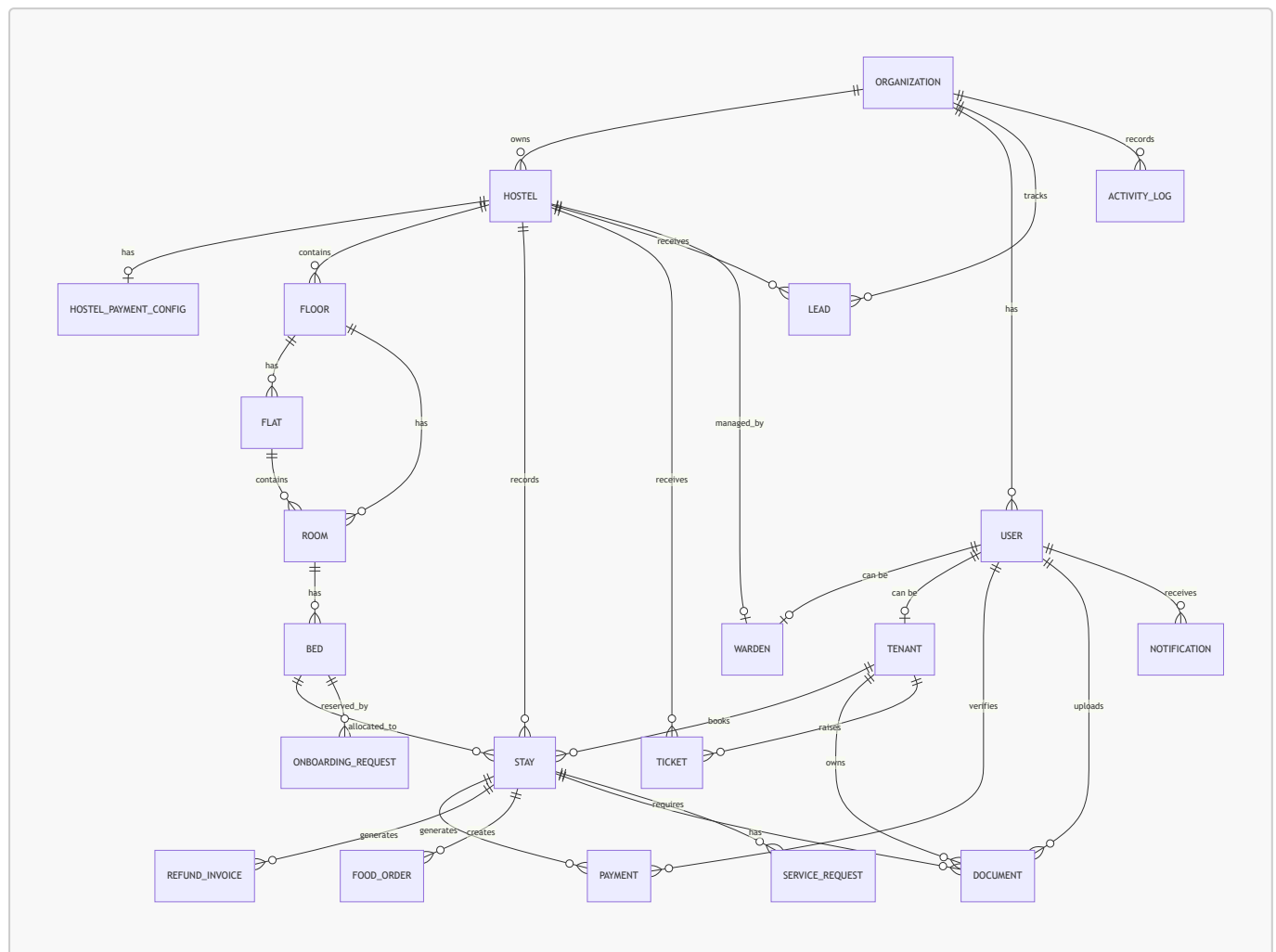
This diagram explains why the Prisma connection pool pattern used in our architecture is architecturally superior to splitting routes across Lambda functions — applicable whether running on EC2 (Phase 1) or ECS Fargate (Phase 2).



Comparison	Lambda Approach	Container Monolith (Our Choice)
DB connections per 50 concurrent req	50 new connections	10 pooled, reused connections
Cold start latency	1–3 seconds per request	0ms — always warm
Cost at scale	Pay per invocation	Predictable flat monthly rate
Next.js compatibility	Requires complex workarounds	Native, first-class support

7. Full Database Schema & Architecture

7.1 Entity-Relationship Diagram



7.2 Deep-Dive Table Architecture

7.2.1 **Organization** — Multi-Tenant Root

The root of the entire SaaS data model. Every record in the system traces back to **organizationId**. This guarantees strict tenant isolation — Organization A can never see Organization B's hostels, tenants, or payments.

7.2.2 **User** + **Warden** + **Tenant** — Role Separation

A single **User** holds credentials and a **UserRole** enum (**MAIN_ADMIN**, **WARDEN**, **TENANT**). Role-specific data lives in dedicated **Warden** and **Tenant** child tables linked 1-to-1. Avoids the "God table" anti-pattern where one table has 50 nullable columns irrelevant to most roles.

7.2.3 Building Hierarchy — **Hostel** → **Floor** → **Flat** → **Room** → **Bed**

Five-level recursive hierarchy. **Room.flatId** and **Room.floorId** are both optional (one must be set), enabling the same schema to model both standard dorm layouts and apartment-style hostel layouts without any code

change.

7.2.4 Stay — The Core Operational Model

Links a **Tenant** to a **Bed**. Stores **hostelId** directly (intentional denormalization) to avoid 4-table JOINS on every dashboard query. **StayStatus** enum acts as a strict finite-state machine: **ONBOARDING_PENDING** → **ACTIVE** → **CHECKED_OUT**.

Financial precision: All monetary values stored as **Paise (integer)** — never floating-point rupees. This prevents IEEE 754 floating-point rounding errors in all financial calculations.

7.2.5 Payment + RefundInvoice — Financial Audit Trail

Auto-incrementing **receiptNumber** is sequential and tamper-evident. Each Payment holds a 1-to-1 reference to an S3 document (UPI screenshot). **verifiedByUserId** and **verifiedAt** create a complete, auditable verification chain.

7.2.6 Document — Polymorphic Storage

A Document can belong to a **Tenant** (KYC: Aadhaar, PAN) or a **Stay** (receipts, registration forms). **storagePath** stores the S3 object key only — never the full URL. Pre-signed download URLs are generated at request time, ensuring no document is permanently publicly accessible.

7.2.7 ActivityLog — Complete Audit Trail

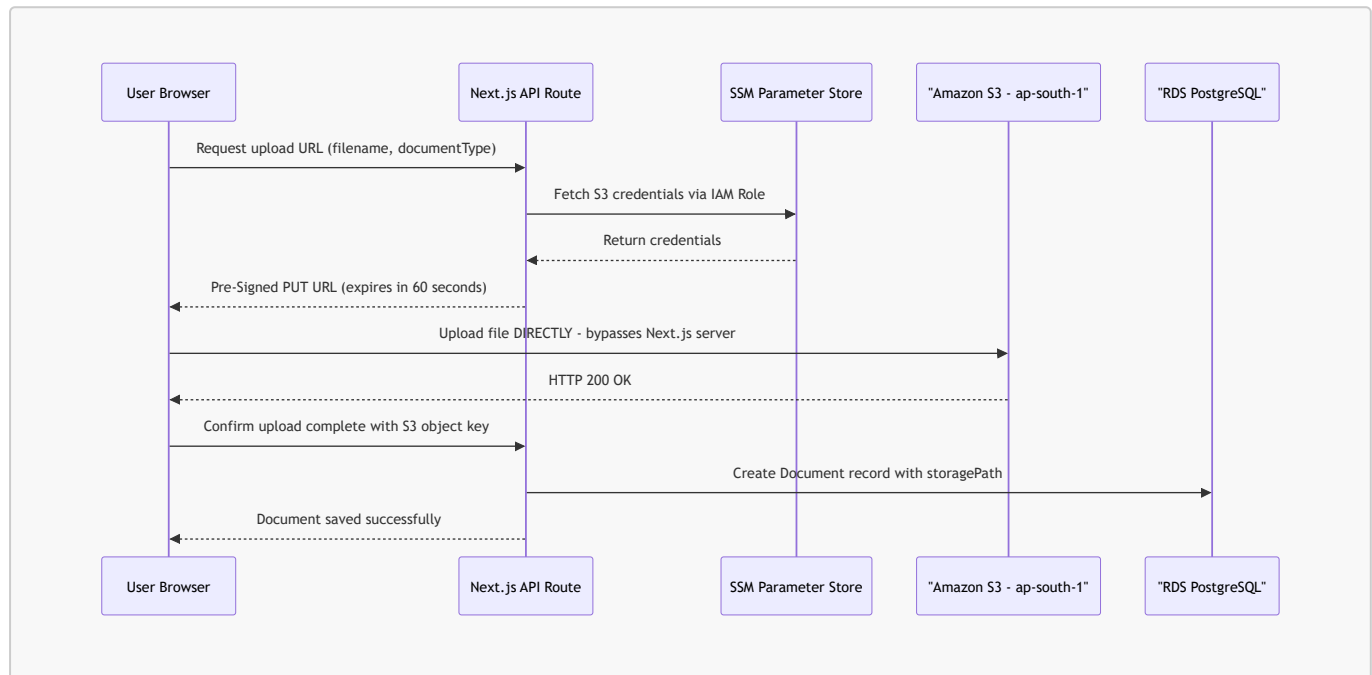
Every significant event writes to **ActivityLog** with **actorName** and **subjectName** stored as denormalized strings. If a User is later deleted, their name is preserved in the audit log permanently — full historical traceability is maintained.

8. Cloud Storage & Document Management

8.1 Why S3 Over Database BLOBs

Storing file contents directly in PostgreSQL is a severe anti-pattern: it balloons the RDS volume, cripples backup/restore times, and forces all file traffic through the database connection pool. Amazon S3 is purpose-built for binary object storage — infinitely scalable, independently priced, and decoupled from the database.

8.2 Pre-Signed URL Upload Flow



Key benefit: The file bytes never pass through the Next.js server. The EC2 container's RAM and CPU are completely unaffected by large file uploads.

8.3 S3 Bucket Layout

```

hostel-prod-documents/ (ap-south-1)
├── kyc/{tenantId}/
│   ├── aadhaar.jpg
│   ├── pan.jpg
│   └── passport_photo.jpg
├── payments/{stayId}/{paymentId}/
│   └── upi_screenshot.jpg
└── pdfs/{stayId}/
    ├── registration_form.pdf
    └── receipt_{receiptNumber}.pdf
  
```

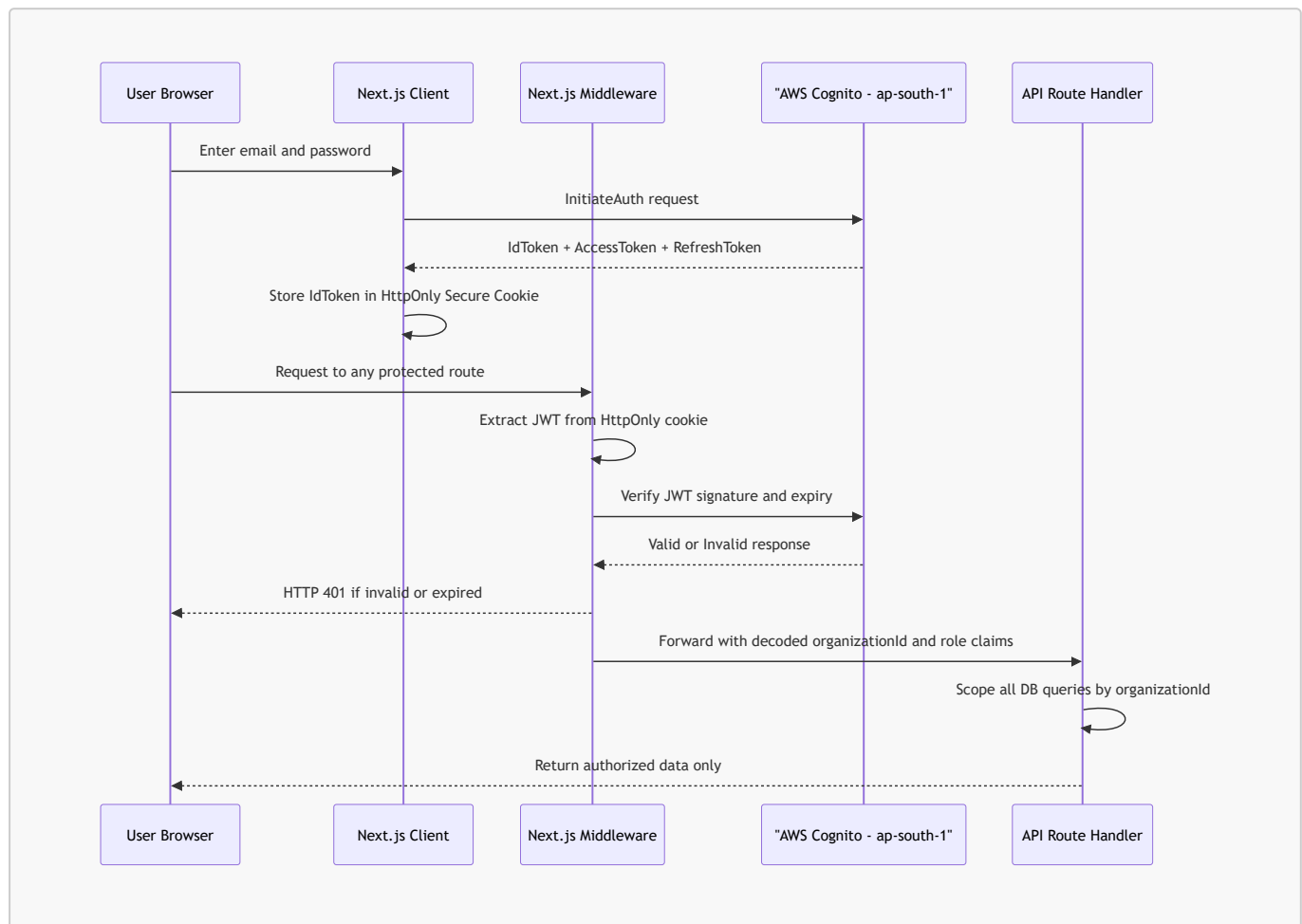
Security controls:

1. Block All Public Access is enabled — no document is directly accessible via URL.
2. Upload Pre-Signed URLs expire in 60 seconds. Download URLs expire in 15 minutes.

3. S3 Versioning is enabled — accidentally overwritten documents are recoverable from version history.

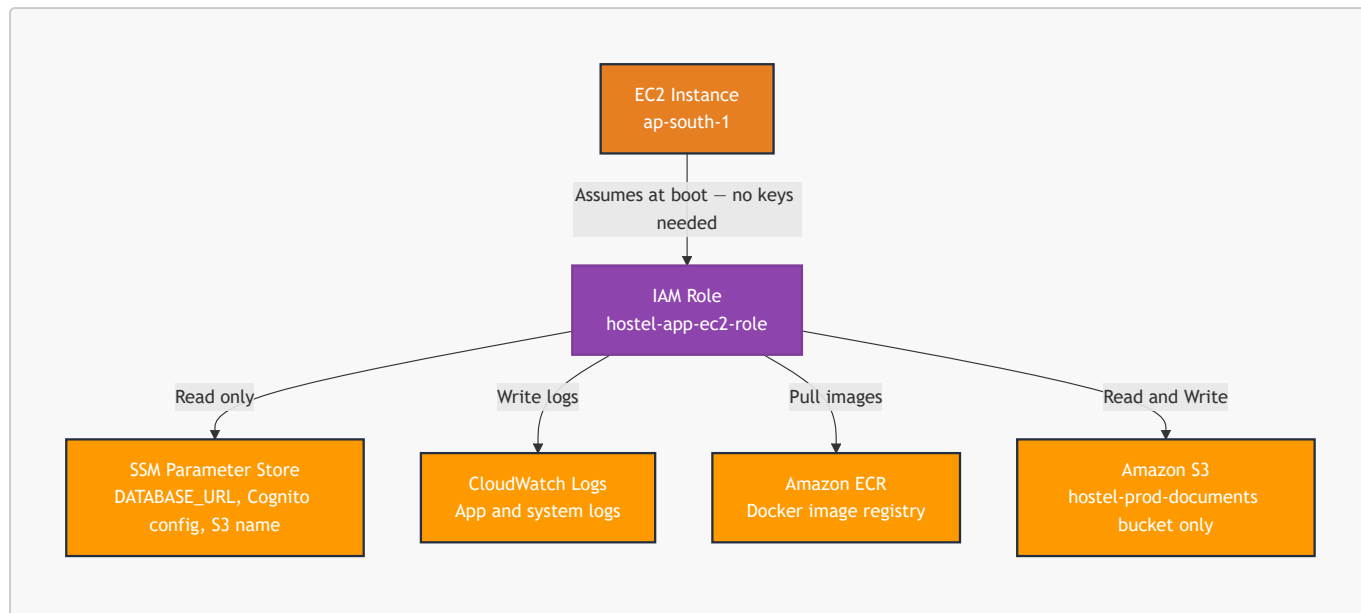
9. Identity, Access Management & Secrets

9.1 JWT Authentication Flow

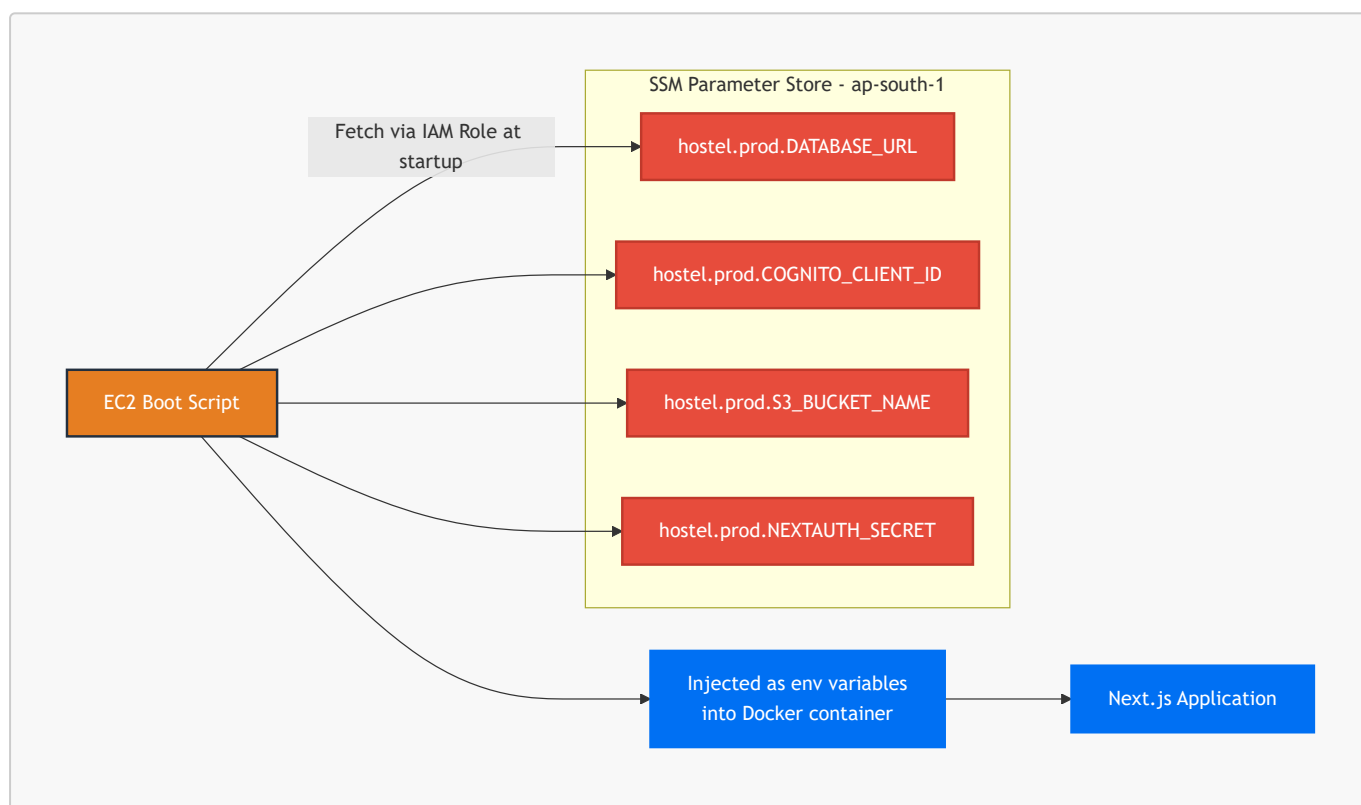


9.2 IAM Task Role — Zero Hardcoded Credentials

No AWS access keys exist anywhere in the codebase, Docker image, or environment files. The EC2 instance is assigned an IAM Role at launch. The AWS SDK automatically inherits credentials from the EC2 instance metadata service.



9.3 Secrets Management — SSM Parameter Store

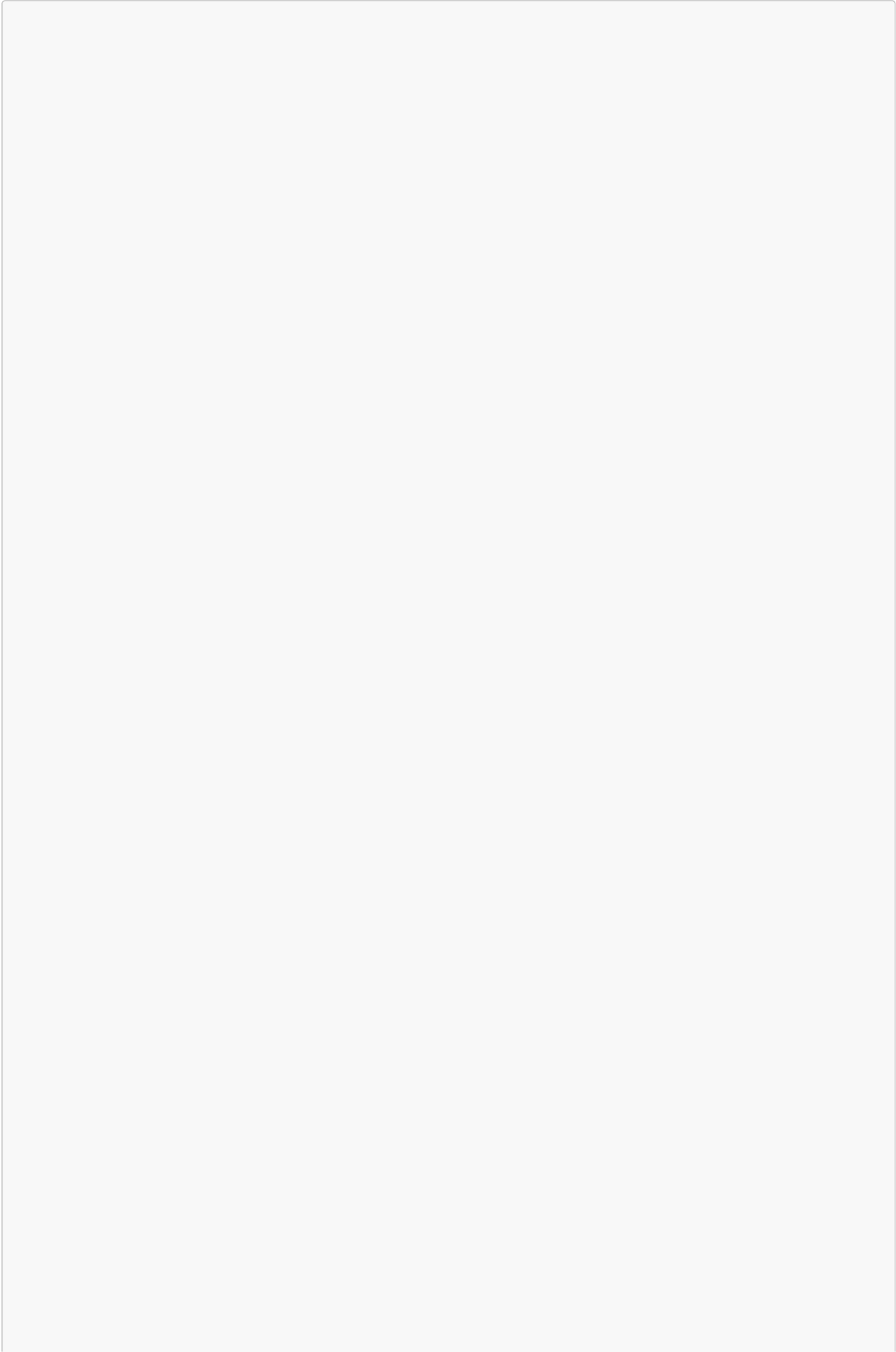


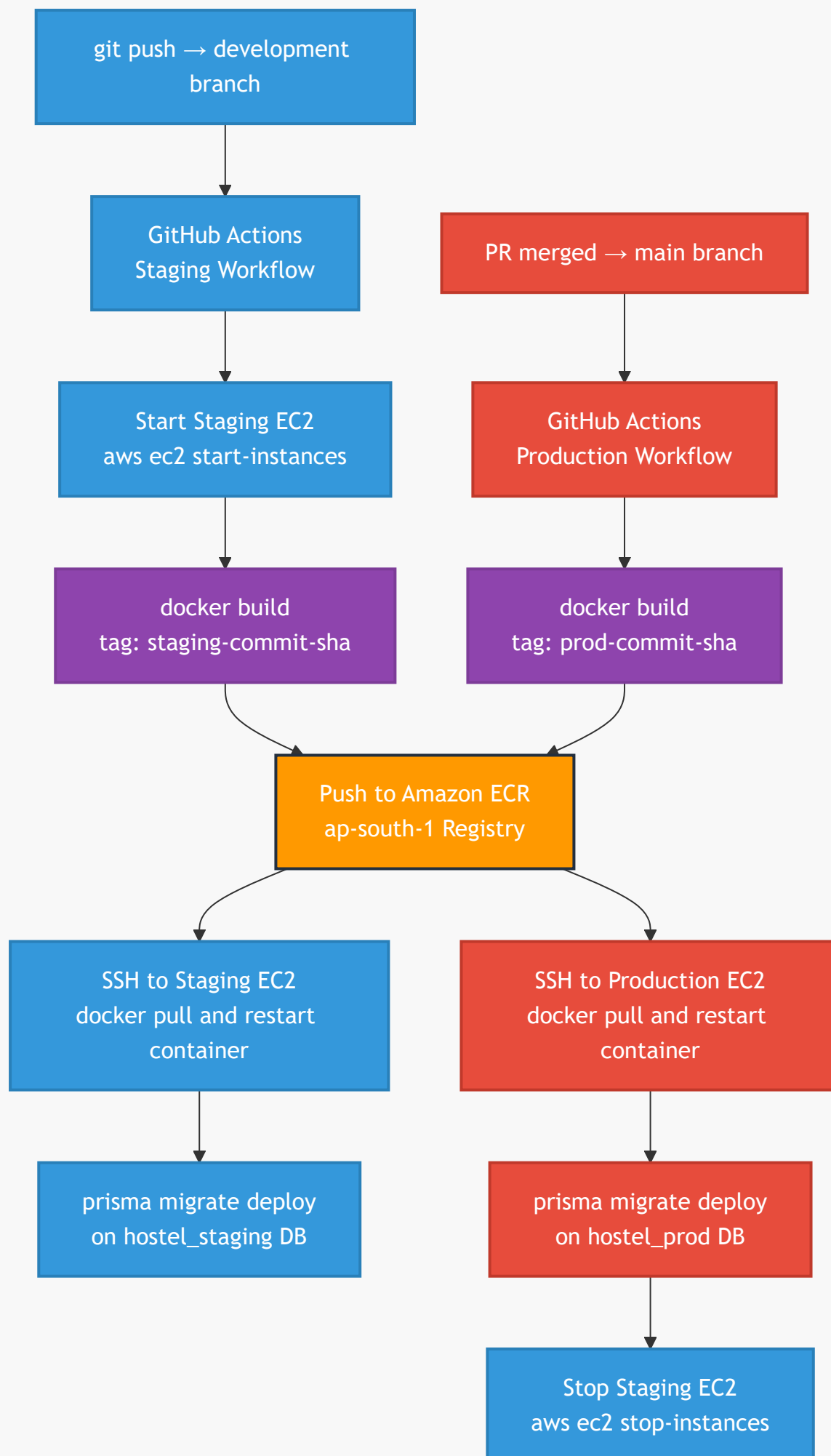
Why SSM over .env files:

- If EC2 is compromised, SSM secrets are not accessible without IAM role permissions
- Secret rotation (e.g., new DB password) requires updating one SSM value — no redeployment
- Every secret access is logged in AWS CloudTrail for compliance auditing

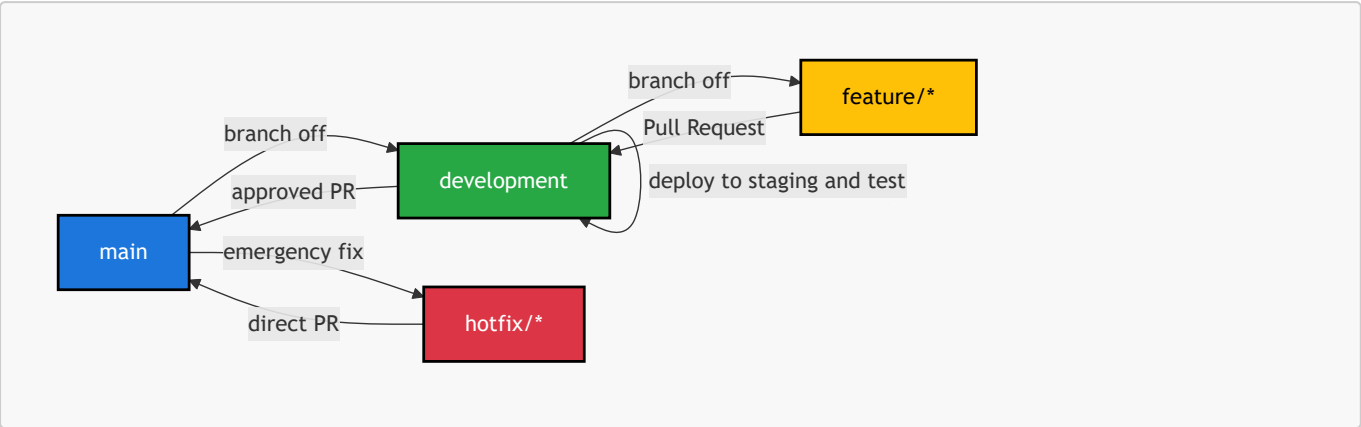
10. CI/CD Pipeline & Deployment

10.1 Complete Pipeline with ECR — Dual Environment





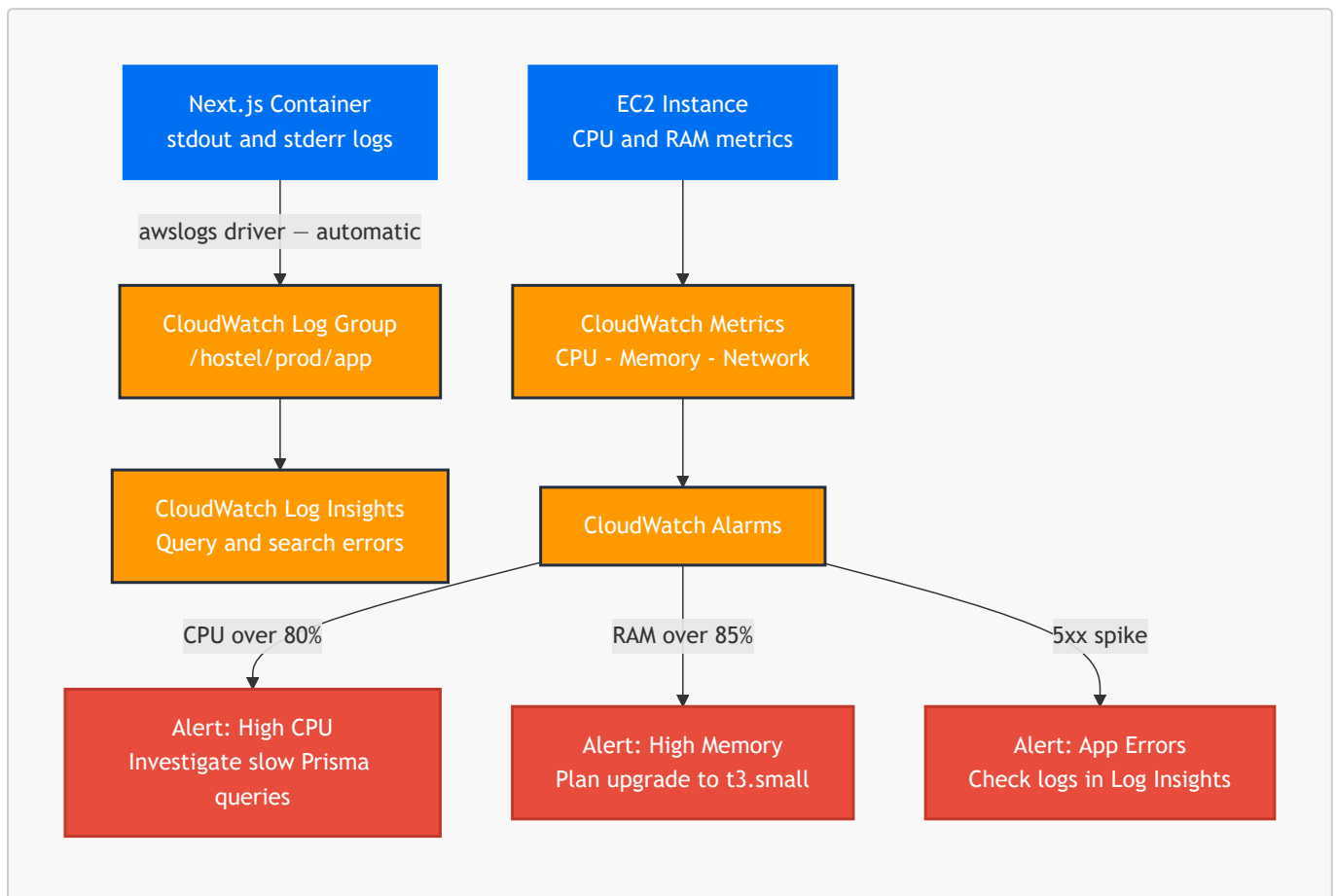
10.2 Git Branching Strategy



Branch	Deploys To	Rules
main	Production EC2	Requires PR + review. No direct push.
development	Staging EC2 (auto-start)	Developer direct push allowed
feature/*	Local machine only	No cloud deployment
hotfix/*	Production (emergency path)	Requires PR directly to main

11. Observability & Monitoring

11.1 CloudWatch Integration



11.2 Log Retention Policy

Log Source	CloudWatch Group	Retention
Next.js app errors	<code>/hostel/prod/app</code>	30 days
Prisma query errors	<code>/hostel/prod/app</code>	30 days
EC2 system logs	<code>/hostel/prod/system</code>	7 days
Docker container logs	<code>/hostel/prod/docker</code>	7 days

All within the 5GB/month permanent free tier. Zero cost.

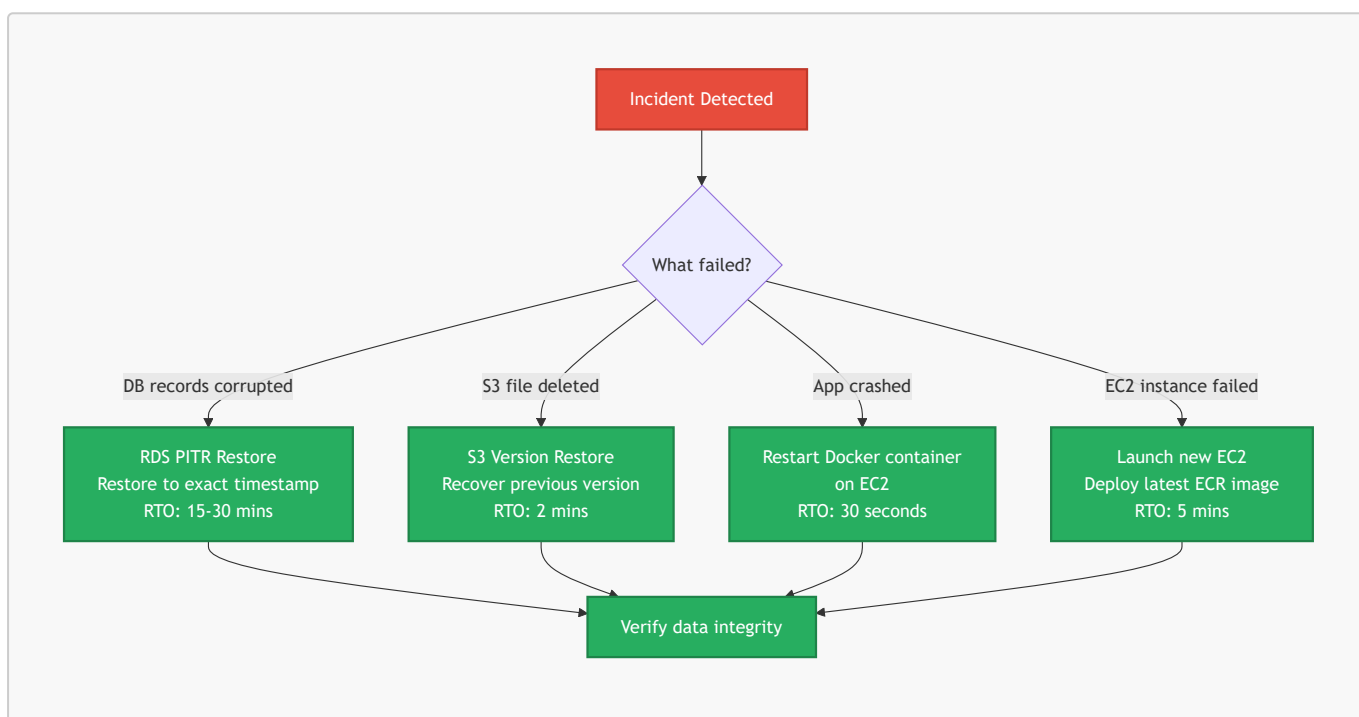
12. Disaster Recovery & Backup

12.1 RDS Point-In-Time Recovery (PITR)

RDS automated backups with PITR are enabled. The database can be restored to any exact second within the last 35 days.

Example scenario: A warden runs a bulk operation that corrupts 500 tenant records at 3:00 PM. The database is restored to its state at 2:55 PM. Zero data loss beyond 5 minutes. Recovery Time Objective (RTO): approximately 15–30 minutes.

12.2 Recovery Decision Tree



13. Cost Optimization & MVP Budget

13.1 Phase 1 MVP — AWS Free Tier (ap-south-1, Year 1)

AWS Service	Free Tier Limit	Expected Usage	Monthly Cost
EC2 t3.micro (Production)	750 hrs/month combined	720 hrs (24/7)	₹0
EC2 t3.micro (Staging)	(shared 750 hr pool)	~20 hrs (on-demand)	₹0
RDS t3.micro (1 instance)	750 hrs/month	720 hrs (24/7)	₹0
RDS Storage	20 GB SSD	~3–5 GB	₹0
Amazon S3	5 GB + 20k GETs/month	~1–2 GB	₹0
AWS Cognito	50,000 MAU (permanent)	< 500 MAU	₹0
CloudWatch Logs	5 GB/month (permanent)	< 1 GB	₹0
CloudFront	1 TB data transfer/month	< 10 GB	₹0
ACM SSL Certificate	Free (permanent)	1 wildcard cert	₹0
SSM Parameter Store	10,000 API calls/month	< 100 calls	₹0
Amazon ECR	500 MB storage/month	< 200 MB	₹0
Route 53	Not free	1 hosted zone	~₹42/month
TOTAL			~₹42/month

[!NOTE] Route 53 at \$0.50/month (~₹42) is the only unavoidable cost. Every other service stays within the AWS Free Tier limits for 12 months at MVP scale.

13.2 Phase Migration Triggers

Trigger Condition	Current Phase	Upgrade To	New Infra Cost
RAM consistently > 80%	Phase 1 EC2 t3.micro	EC2 t3.small (2GB RAM)	~₹2,500/month
15+ active hostels	Phase 1	Phase 1.5 (t3.small)	~₹2,500/month
30+ active hostels	Phase 1.5	Phase 2 (ECS Fargate)	~₹10,300/month
100+ active hostels	Phase 2	Phase 2 scaled (bigger RDS)	~₹18,000/month

13.3 Phase 2 Scale Budget (30+ Hostels)

AWS Service	Configuration	Monthly Cost (ap-south-1)
ECS Fargate	0.5 vCPU, 1GB RAM, 2 tasks	~₹2,500
RDS t3.small	Single-AZ, 50GB SSD	~₹2,800

AWS Service	Configuration	Monthly Cost (ap-south-1)
Application Load Balancer	1 ALB	~₹1,400
CloudFront	100GB data transfer	~₹700
CloudWatch + ECR + SSM	Combined	~₹500
ElastiCache t3.micro	Redis (if needed at scale)	~₹1,200
Route 53	1 hosted zone	~₹42
RDS Multi-AZ (optional)	Add-on for HA	+₹1,200
TOTAL		~₹10,300/month

14. Revenue & Infrastructure Forecast

[!IMPORTANT] **This section contains hypothetical projections only.** The pricing figures (₹1,000/hostel/month) are used purely to illustrate the relationship between customer growth, revenue, and infrastructure cost. Actual pricing strategy is not fixed and will be determined based on market research, feature completeness, and competitive positioning.

14.1 The SaaS Infrastructure Economics Principle

Infrastructure costs in a SaaS product scale **sub-linearly** — meaning that doubling the number of customers does not double the infrastructure cost. A single ECS Fargate cluster or EC2 instance that handles 30 hostels can handle 100 hostels with little to no cost increase. This is the fundamental reason SaaS margins improve dramatically at scale.

14.2 Revenue vs Infrastructure — Hypothetical at ₹1,000/Hostel/Month

Phase	Active Hostels	Monthly Revenue	Infra Cost	Gross Profit	Margin
Phase 1 — Free Tier	5	₹5,000	₹42	₹4,958	99%
Phase 1 — Growing	9	₹9,000	₹42	₹8,958	99%
Phase 1 — Full	15	₹15,000	₹42	₹14,958	99%
Phase 1.5 — Upgrade	20	₹20,000	₹2,500	₹17,500	87%
Phase 2 — Migration	30	₹30,000	₹10,300	₹19,700	65%
Phase 2 — Sweet Spot	50	₹50,000	₹10,300	₹39,700	79%
Phase 2 — Scale	100	₹1,00,000	₹10,300	₹89,700	89%
Phase 2 — Enterprise	150	₹1,50,000	₹10,300	₹1,39,700	93%

[!NOTE] After crossing 30 hostels and migrating to Phase 2, every additional hostel added costs approximately ₹0 more in infrastructure. The ₹10,300/month Phase 2 infra serves 30 hostels and 150 hostels almost identically — demonstrating classic SaaS margin expansion at scale.

14.3 The Price Increase Strategy

The platform is designed to launch at a competitive introductory price and increase as the product matures with additional features.

Product Stage	Planned Feature Set	Justifiable Price
MVP Launch	Core hostel management, payments, KYC	₹1,000/hostel/month
Growth Phase	Food billing, task management, service requests, analytics	₹2,000/hostel/month
Mature Product	WhatsApp bot, tenant mobile app, multi-hostel reporting	₹3,000+/hostel/month

14.4 Revised Forecast at Higher Pricing

Active Hostels	At ₹2,000/hostel	Infra Cost	Margin
30	₹60,000	₹10,300	82%
50	₹1,00,000	₹10,300	89%
100	₹2,00,000	₹10,300	94%

[!TIP] The migration from Phase 1 to Phase 2 infra (₹42 → ₹10,300) should only happen when revenue has already grown enough to sustain it comfortably. At 30 hostels $\times$ ₹1,000 = ₹30,000 revenue, the infra is 34% of revenue which is acceptable. At 30 hostels $\times$ ₹2,000 = ₹60,000, infra is only 17% — a healthy SaaS ratio.

15. Exhaustive API Documentation

All routes below run inside the single Next.js Docker container on EC2 (ap-south-1). All share one Prisma connection pool against RDS PostgreSQL.

GET/POST/PUT/DELETE /api/admin/activity

- **Domain:** `admin` module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE /api/admin/activity/export

- **Domain:** `admin` module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE /api/admin/beds/[id]

- **Domain:** `admin` module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE /api/admin/dashboard/stats

- **Domain:** `admin` module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE /api/admin/flats

- **Domain:** `admin` module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE `/api/admin/flats/[id]`

- **Domain:** `admin` module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE `/api/admin/floors`

- **Domain:** `admin` module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE `/api/admin/floors/[id]`

- **Domain:** `admin` module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE `/api/admin/hostels`

- **Domain:** `admin` module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE `/api/admin/hostels/[id]/payment-config`

- **Domain:** `admin` module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE `/api/admin/hostels/[id]/warden`

- **Domain:** `admin` module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE `/api/admin/locations`

- **Domain:** `admin` module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE `/api/admin/onboards`

- **Domain:** `admin` module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE `/api/admin/onboards/[id]/cancel`

- **Domain:** `admin` module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE `/api/admin/rooms`

- **Domain:** `admin` module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE `/api/admin/rooms/[id]`

- **Domain:** `admin` module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE `/api/admin/tickets`

- **Domain:** `admin` module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE `/api/admin/tickets/[id]/comments`

- **Domain:** `admin` module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE `/api/admin/users`

- **Domain:** `admin` module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE `/api/admin/users/[id]/reset-password`

- **Domain:** `admin` module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE /api/admin/wardens

- **Domain:** `admin` module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE /api/admin/wardens/[id]

- **Domain:** `admin` module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE /api/admin/wardens/[id]/reset-password

- **Domain:** `admin` module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE /api/auth/login

- **Domain:** `auth` module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE /api/auth/logout

- **Domain:** `auth` module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE `/api/auth/reset-password`

- **Domain:** `auth` module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE `/api/auth/set-password`

- **Domain:** `auth` module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE `/api/hostel-structure/[hostelId]`

- **Domain:** `hostel-structure` module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE `/api/hostel-structure/mine`

- **Domain:** `hostel-structure` module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE `/api/internal/auth-check`

- **Domain:** `internal` module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE /api/notifications

- **Domain:** `notifications` module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE /api/notifications/[id]

- **Domain:** `notifications` module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE /api/pdf/download/[documentId]

- **Domain:** `pdf` module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE /api/pdf/receipt/[paymentId]

- **Domain:** `pdf` module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE /api/pdf/refund-invoice/[refundInvoiceId]

- **Domain:** pdf module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE /api/pdf/registration-form/[stayId]

- **Domain:** pdf module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE /api/public/hostels/[id]/payment-config

- **Domain:** public module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE /api/public/onboard-request/[id]

- **Domain:** public module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE /api/public/onboard-request/[id]/register

- **Domain:** public module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE /api/public/onboarding/[id]

- **Domain:** `public` module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE `/api/public/onboarding/[id]/finalize`

- **Domain:** `public` module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE `/api/public/onboarding/[id]/progress`

- **Domain:** `public` module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE `/api/public/onboarding/[id]/validate`

- **Domain:** `public` module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE `/api/tenant/food-orders`

- **Domain:** `tenant` module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE `/api/tenant/payment/screenshot`

- **Domain:** `tenant` module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE `/api/tenant/service-requests/[id]/payment`

- **Domain:** `tenant` module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE `/api/tenant/settings`

- **Domain:** `tenant` module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE `/api/tenant/settings/password`

- **Domain:** `tenant` module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE `/api/tenant/stay`

- **Domain:** `tenant` module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE `/api/tenant/tickets`

- **Domain:** `tenant` module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE /api/tickets/[id]/comments

- **Domain:** `tickets` module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE /api/warden/action-counts

- **Domain:** `warden` module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE /api/warden/activity

- **Domain:** `warden` module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE /api/warden/activity/export

- **Domain:** `warden` module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE /api/warden/beds/[id]/status

- **Domain:** `warden` module
- **Region:** `ap-south-1` (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL `ap-south-1`). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE `/api/warden/beds/available`

- **Domain:** `warden` module
- **Region:** `ap-south-1` (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL `ap-south-1`). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE `/api/warden/dashboard/stats`

- **Domain:** `warden` module
- **Region:** `ap-south-1` (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL `ap-south-1`). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE `/api/warden/food-mark`

- **Domain:** `warden` module
- **Region:** `ap-south-1` (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL `ap-south-1`). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE `/api/warden/food-stats`

- **Domain:** `warden` module
- **Region:** `ap-south-1` (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL `ap-south-1`). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE `/api/warden/food-week`

- **Domain:** `warden` module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE `/api/warden/leads`

- **Domain:** `warden` module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE `/api/warden/leads/[id]`

- **Domain:** `warden` module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE `/api/warden/onboard`

- **Domain:** `warden` module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE `/api/warden/onboarding-requests/[id]/regenerate-password`

- **Domain:** `warden` module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE /api/warden/onboards

- **Domain:** warden module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE /api/warden/onboards/[id]

- **Domain:** warden module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE /api/warden/onboards/[id]/approve

- **Domain:** warden module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE /api/warden/onboards/[id]/payment

- **Domain:** warden module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE /api/warden/onboards/[id]/reject

- **Domain:** warden module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE /api/warden/onboards/[id]/verify

- **Domain:** warden module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE /api/warden/service-requests/[id]/verify

- **Domain:** warden module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE /api/warden/stays/[id]

- **Domain:** warden module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE /api/warden/stays/[id]/early-checkout

- **Domain:** warden module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE /api/warden/stays/[id]/extend

- **Domain:** warden module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE /api/warden/stays/[id]/refund-estimate

- **Domain:** warden module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE /api/warden/stays/[id]/revoke-food

- **Domain:** warden module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE /api/warden/stays/[id]/service-request

- **Domain:** warden module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE /api/warden/stays/natural-checkout

- **Domain:** warden module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE /api/warden/tickets

- **Domain:** warden module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

GET/POST/PUT/DELETE /api/warden/worklists

- **Domain:** warden module
- **Region:** ap-south-1 (Mumbai)
- **Auth:** Cognito JWT validated by Next.js Middleware. HTTP 401 if token missing or expired.
- **DB Layer:** Prisma ORM on shared connection pool (RDS PostgreSQL ap-south-1). Zero new connections per request.
- **Caching:** Read-heavy endpoints use Next.js Data Cache with `revalidateTag` for instant invalidation on mutation.

AWS Cloud Architecture & Infrastructure Design Document — TrueNorth Hostel Management Platform Version 1.6 — Zenxvio Internal Team — Working Draft — ap-south-1 (Mumbai)